Microeconomics

Lecture 2: Consumer Choice

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Outline

- Concept of commodities
- Consumption set
- Walrasian budget set
- Walrasian demand function
- Compensated law of demand

In this chapter consumer demand is modeled with a choice-based approach.

Next chapter models preferences.

Commodities

A commodity is defined by its physical characteristics and any relevant **date, location, or state of the world**.

With L commodity types, a consumption bundle is

$$x = (x_1, \ldots, x_L).$$

Bread today and bread tomorrow are different commodities. So are deliveries at different locations.

For now, assume finitely many commodities traded at known prices.

The consumption set

The set $X \subseteq \mathbb{R}^L$ describes physical feasibility. Our baseline is $X = \mathbb{R}^L_+$.

Definition (Convexity)

X is convex if, for all $x, y \in X$ and $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in X.$$

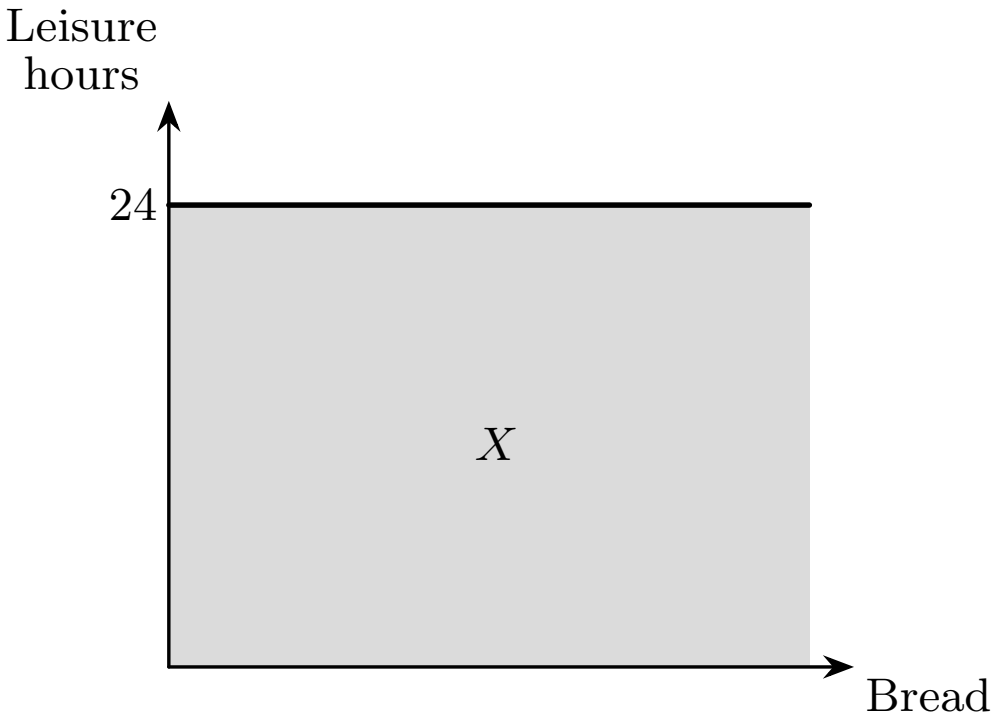
Physical feasibility is distinct from affordability.

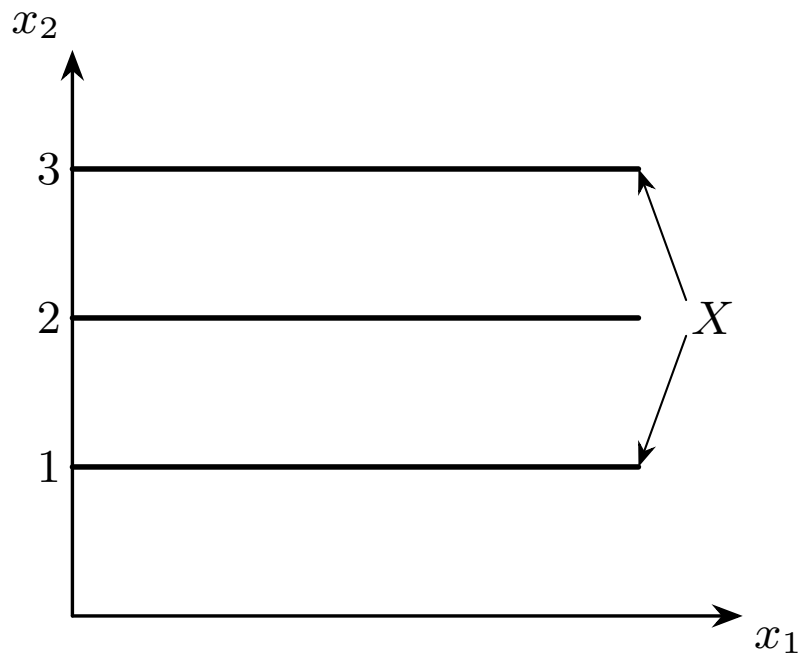
Convexity in microeconomics is usually assumed in three cases:

- Budget sets, Preferences, Production sets.

It is quite likely that it is violated and if that happens the nice properties of competitive markets may not be valid.

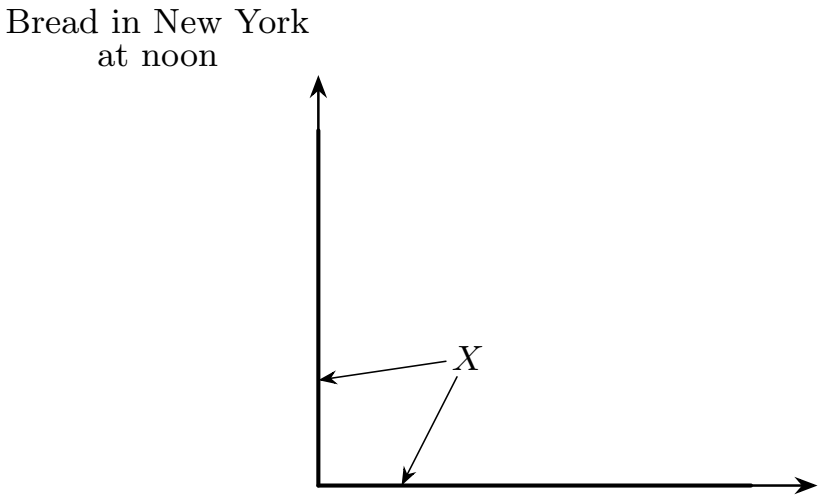
Examples of consumption set



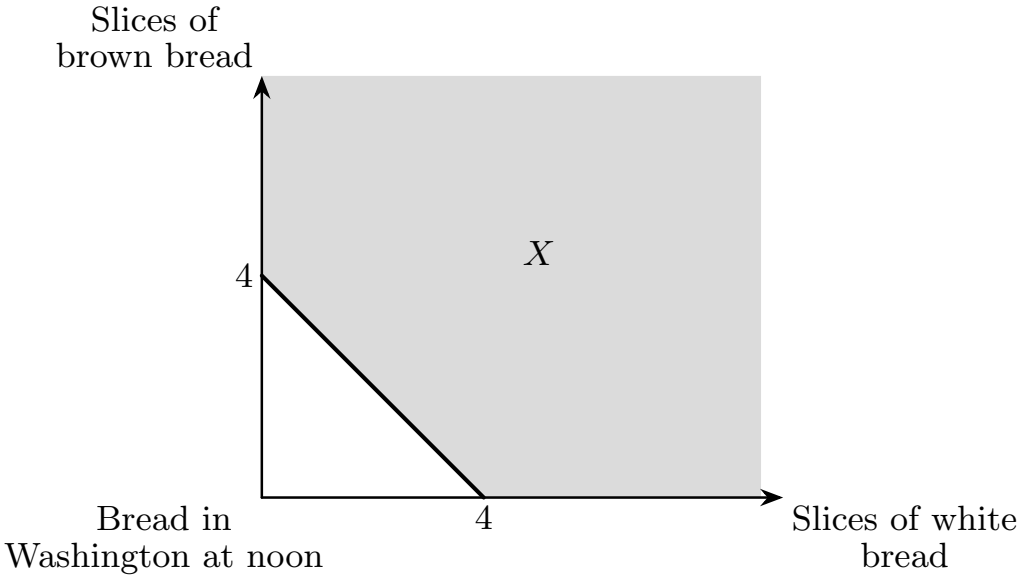


Examples of consumption set

only one good can be consumed



survival needs



Competitive budget

All commodities are traded in the market by prices $p = (p_1, \ldots, p_L)$.

We assume in this chapter

- $\forall l,\ p_l > 0$
- consumers are price-takers.

Total cost of consumption is less than consumer's wealth

$$p.x = p_1 x_1 + \cdots + p_L x_L \leq w$$

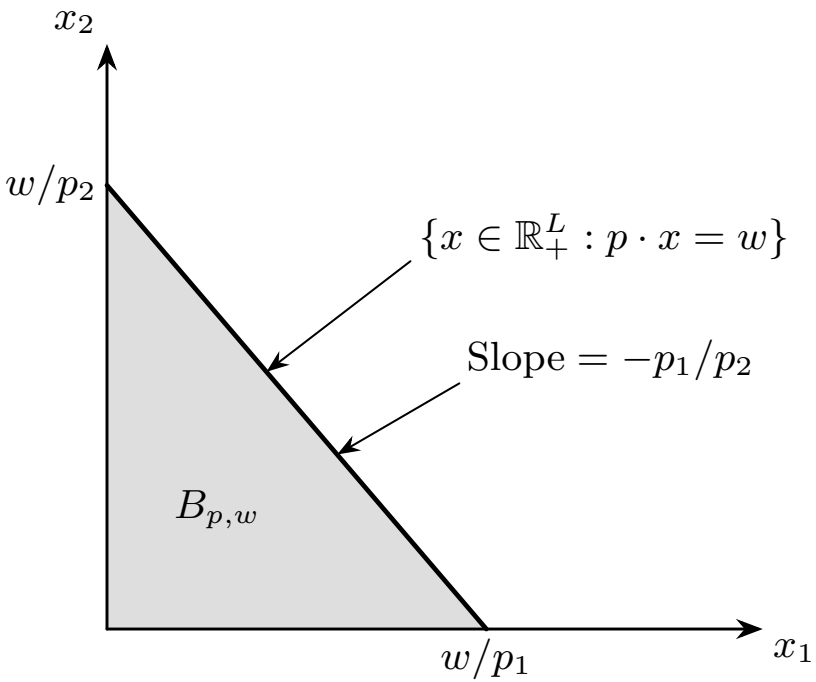
Definition

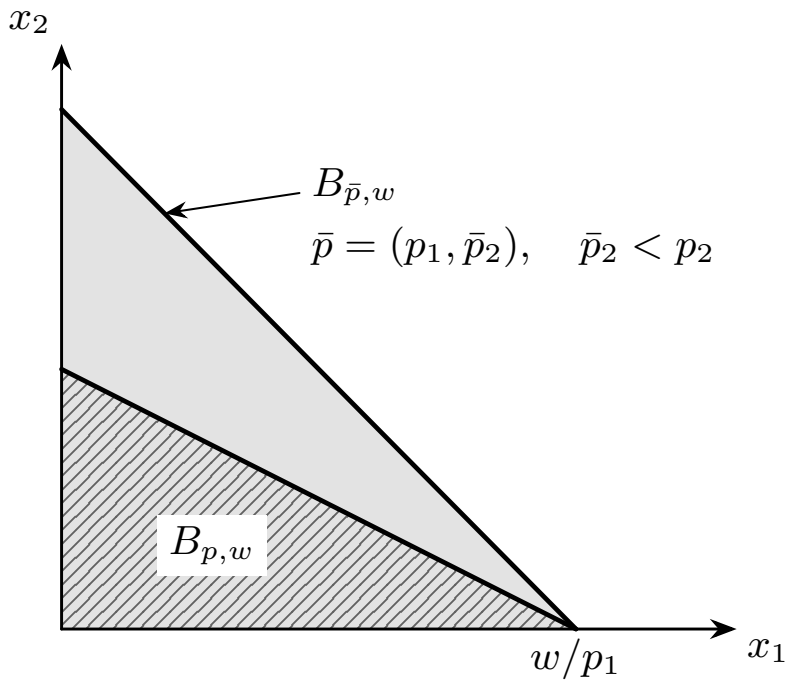
The Walrasian or competitive budget set $B_{p,w} = \{x \in \mathbb{R}^L_+ : p.x \leq w\}$ is the set of all feasible consumption bundles for the consumer who faces market prices p and has wealth w.

Walrasian budget set

Walrasian budget set is a convex set.

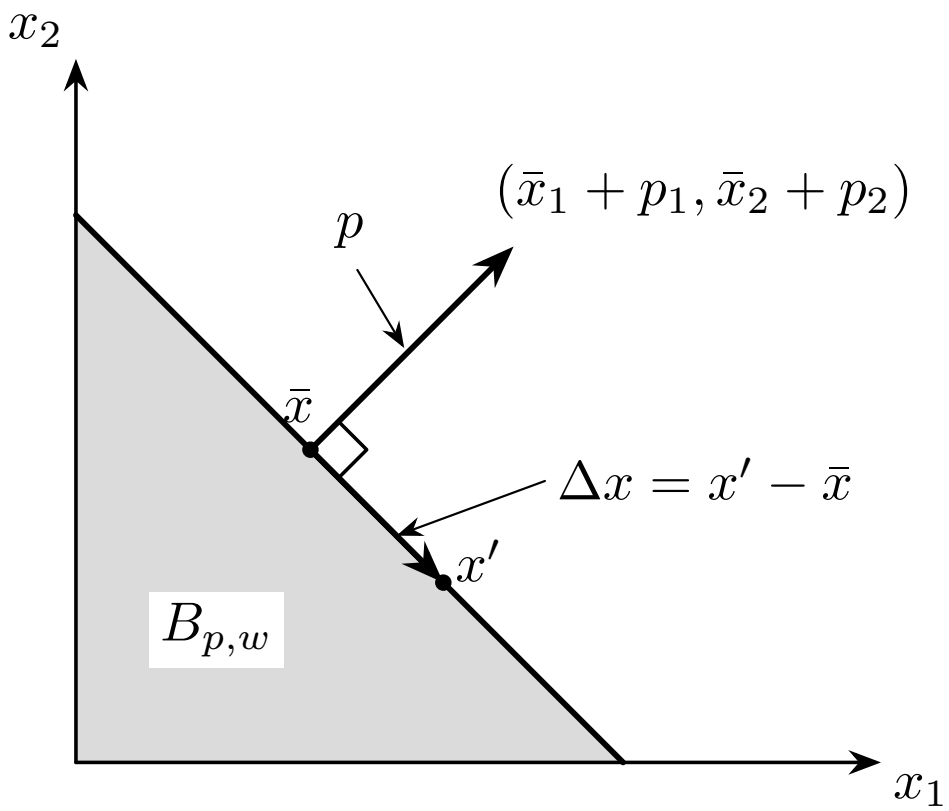
Budget hyperplane: $\{x \in \mathbb{R}^L : p.x = w\}$





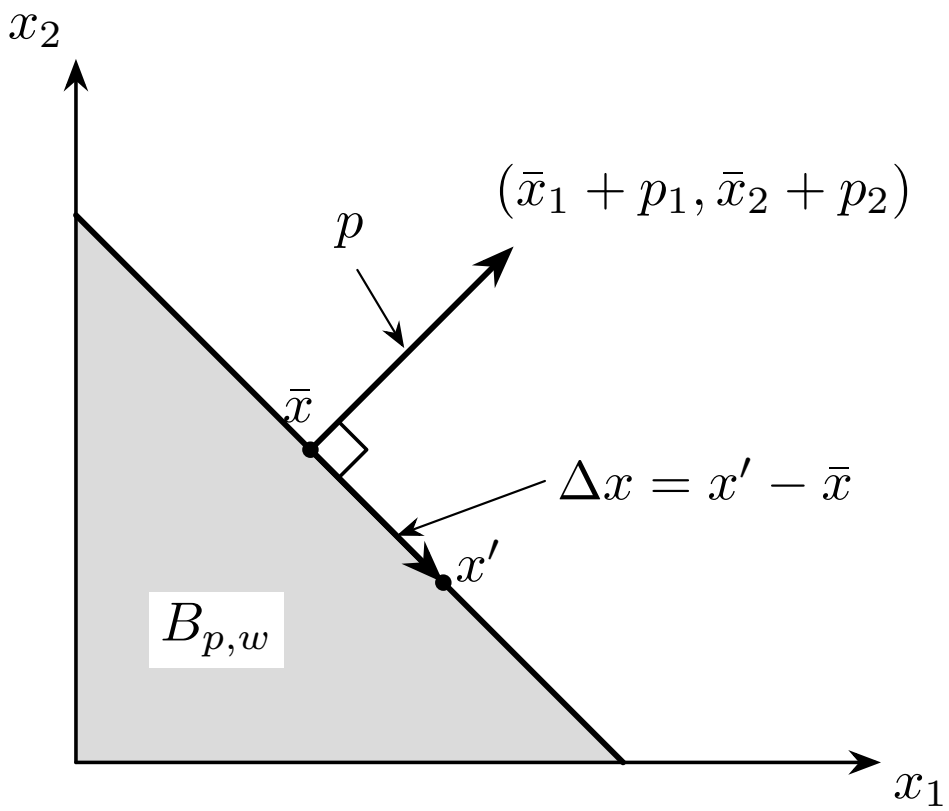
Budget hyperplane

Price vector p is orthogonal to the budget hyperplane. Why?



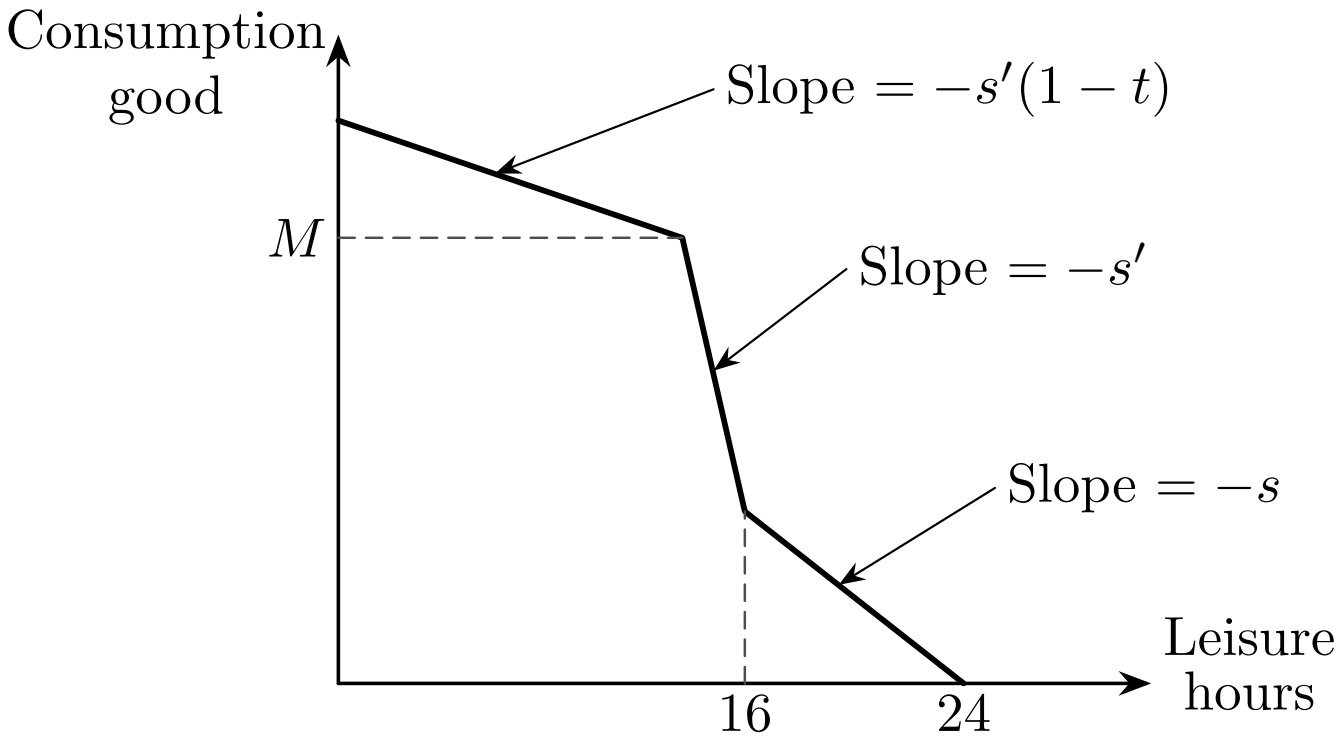
Budget hyperplane

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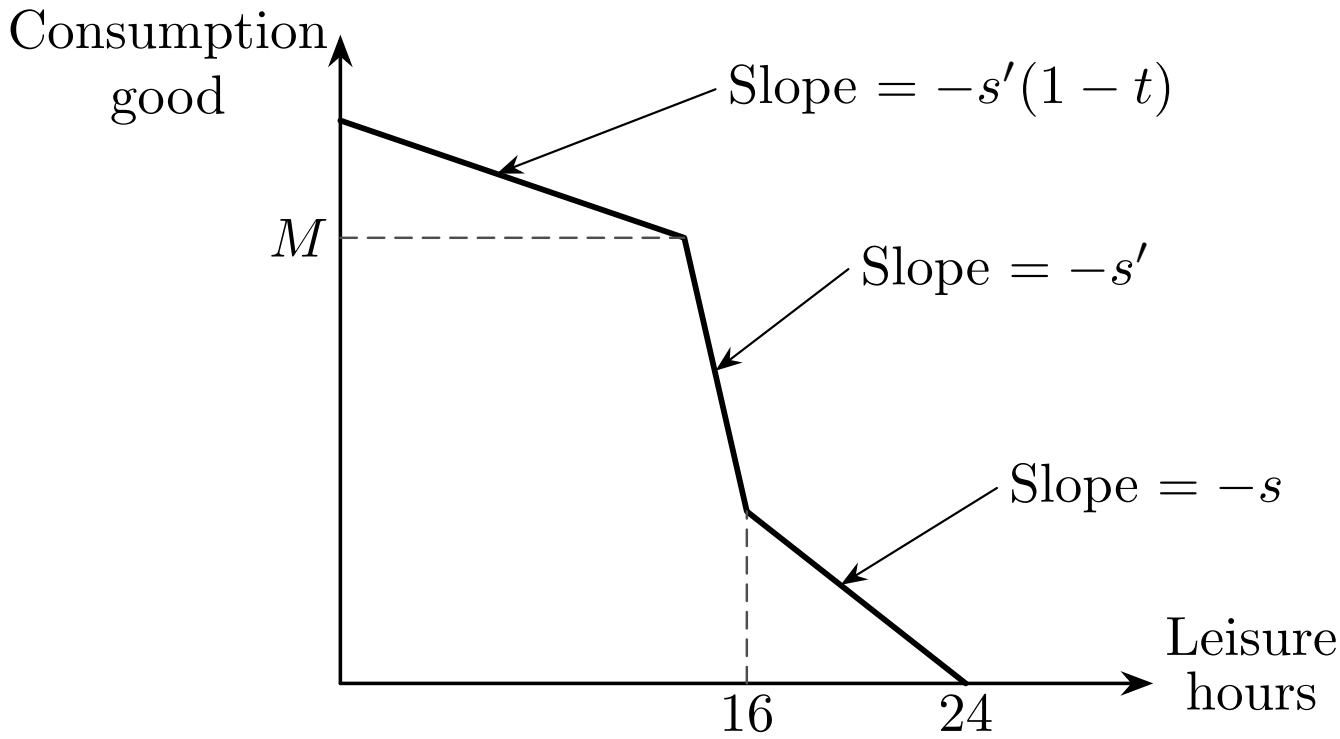


$\forall x'$ on the hyperplane $p.x' = p.\bar{x} = w \Rightarrow p.(x' - \bar{x}) = p.\Delta x = 0.$

Non-convexity of budget sets



Non-convexity of budget sets



Taxes, benefit withdrawal, and quantity discounts can create kinks and nonconvexities.

Walrasian demand

For price p and wealth w, $x(p, w)$ is the set of consumption bundles.

If $x(p, w)$ is always single-valued $x(p, w)$ is a demand function.

Walrasian demand is assumed to be

1. **homogeneous of degree zero**: $x(\alpha p, \alpha w) = x(p, w)$,
2. consumers fully spend their wealth (**Walras' law**): $p.x(p, w) = w$

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The way we proceed here is in line with the choice-based approach:

- Walrasian budget sets: $\mathfrak{B}^W = \{B : p > 0, w > 0\}$
- choice structure: $\{\mathfrak{B}^W, x(p, w)\}$.

In chapter 3, we derive these properties as results of maximization of preferences.

Comparative statistics

Income effect for good l: $\partial x_l / \partial w$

- If $\partial x_l / \partial w \geq 0$, the good l is normal.
- If $\partial x_l / \partial w < 0$, the good l is inferior.

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- If $\partial x_l / \partial p_l < 0$, the good l is regular.
- If $\partial x_l / \partial p_l \geq 0$, the good l is Giffen.

Is a Giffen good always inferior?

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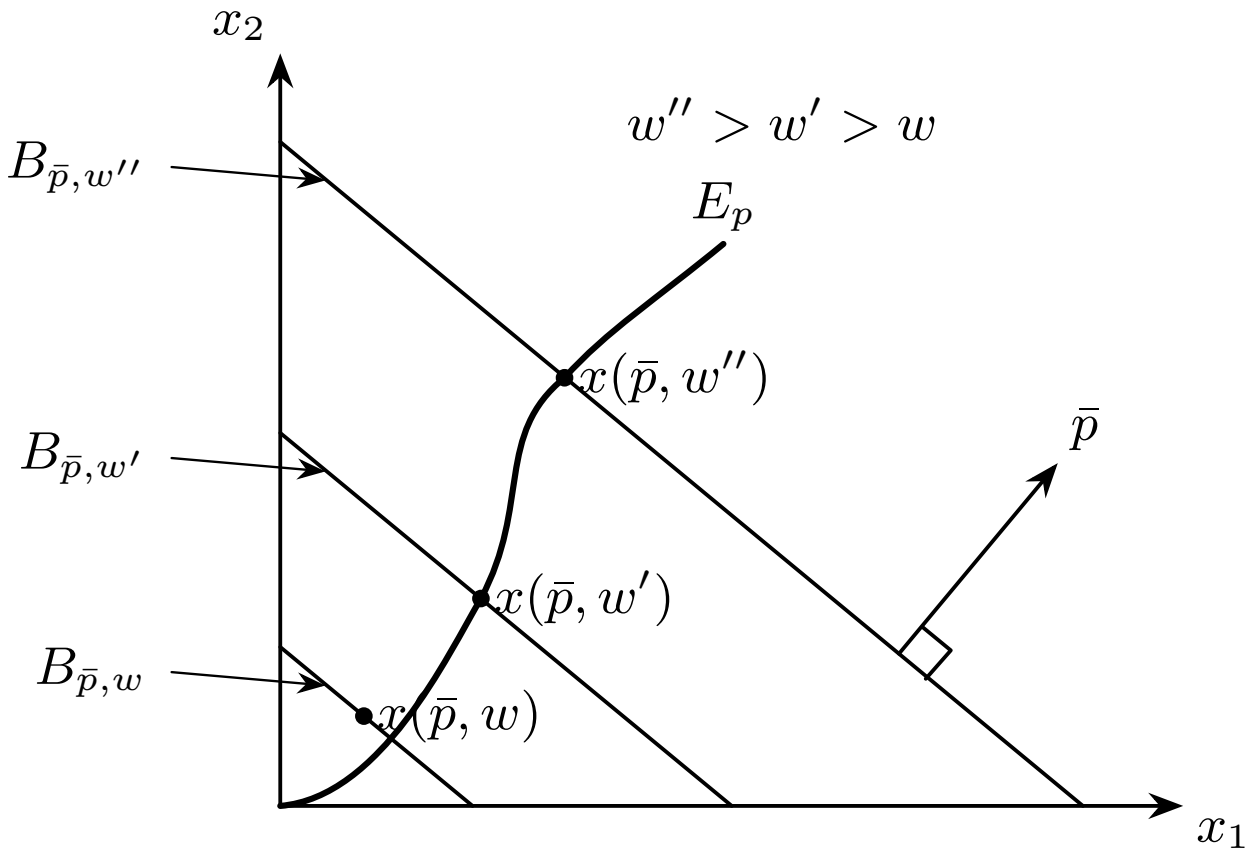
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Price effect of good k on good l: $\partial x_l / \partial p_k$

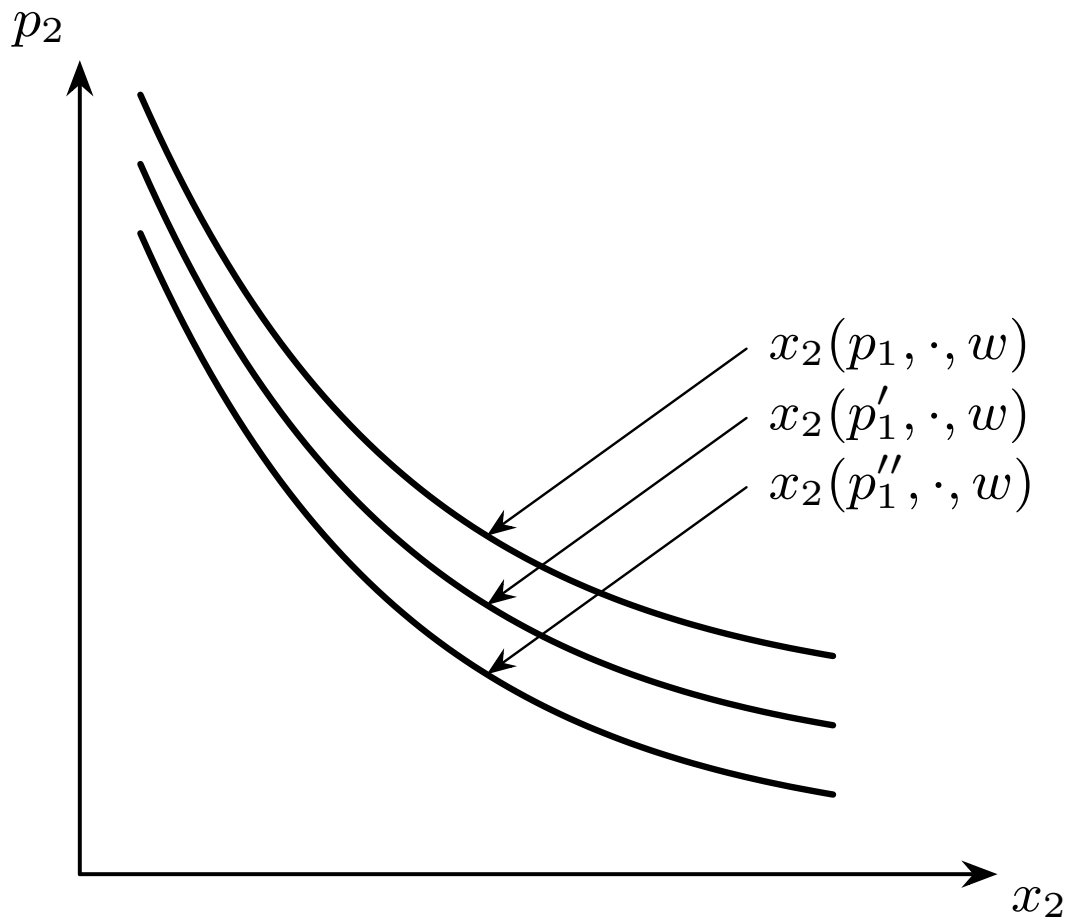
- If $\partial x_l / \partial p_k > 0$, the good l and k are gross substitutes.
- If $\partial x_l / \partial p_k < 0$, the good l and k are gross complement.

Income (wealth) effect



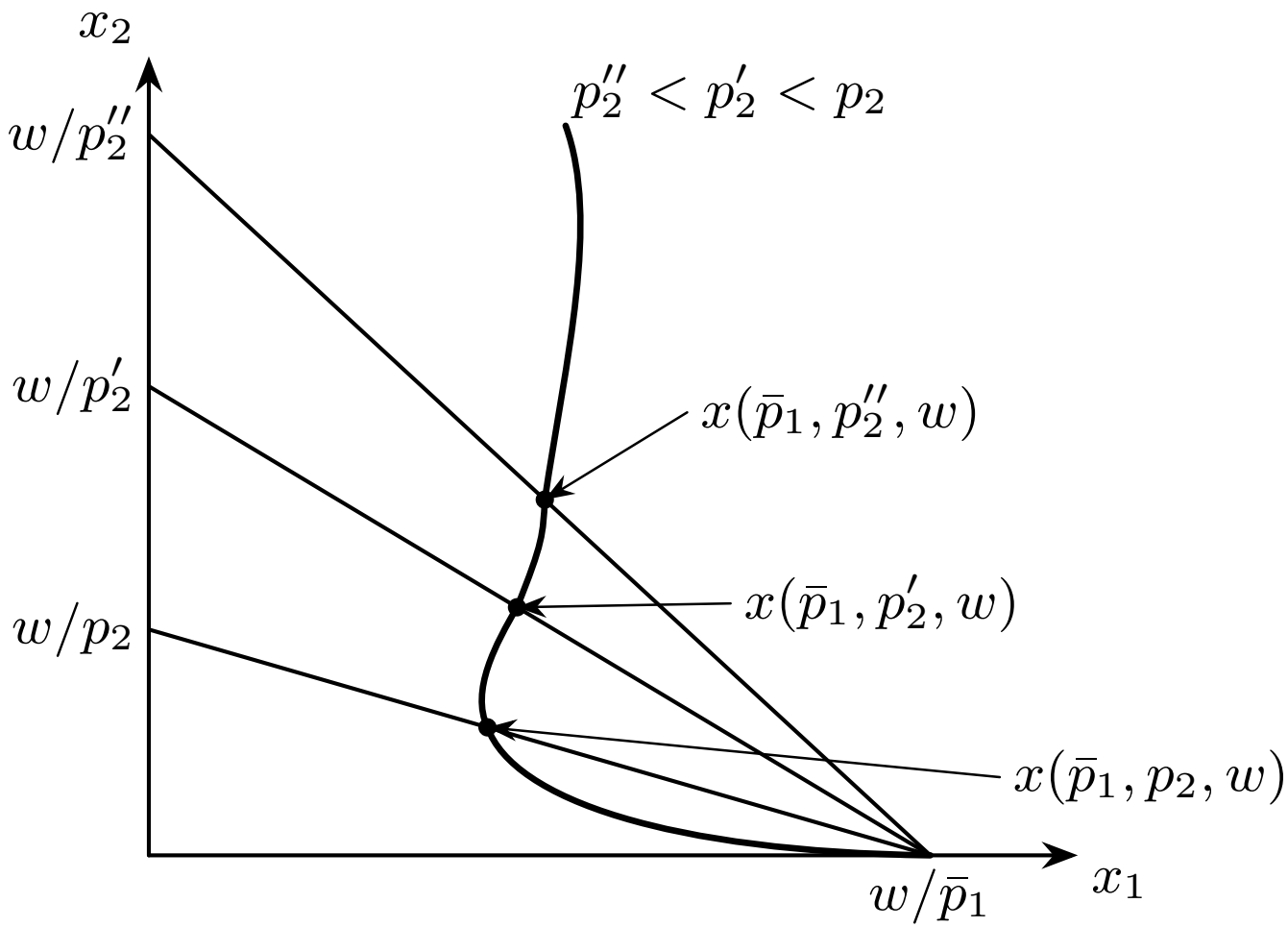
The figure is a wealth expansion path in commodity space. An Engel curve $x(\bar{p}, w)$ plots one x_ℓ against w.

Price effect



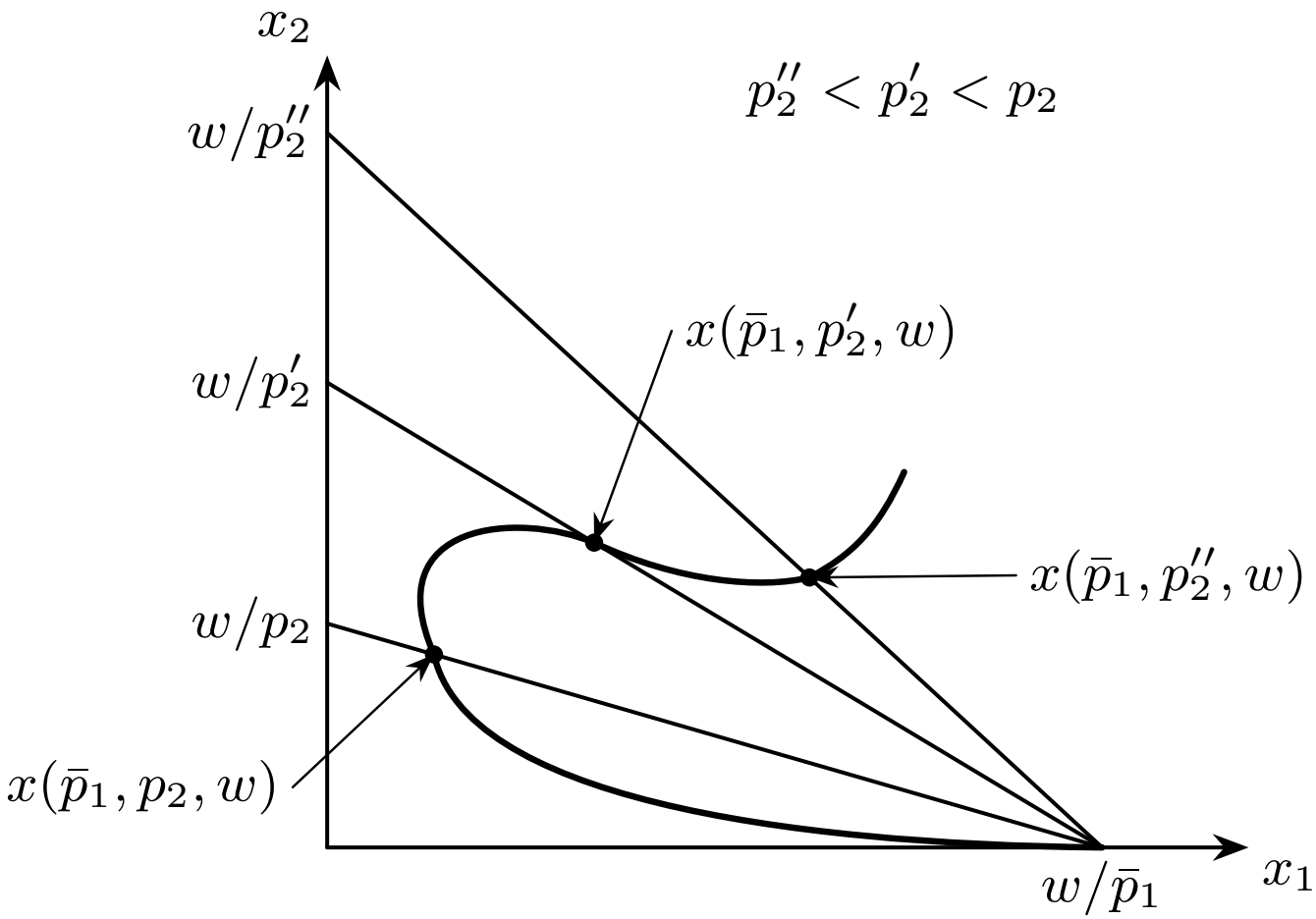
A movement along x_2 changes p_2; a shift changes another price or wealth.

Offer curve



The price offer curve tracks the chosen bundle as one price changes.

An offer curve can bend backward



Identify which coordinate increases or decreases as p_2 falls.

Implication of homogeneity of demand

Demand function is homogeneous of degree zero: $x(\alpha p, \alpha w) = x(p, w)$.

Derivation w.r.t. α yields:

$$\sum_{k=1}^{L} \frac{\partial x_l(p,w)}{\partial p_k} p_k + \frac{\partial x_l(p,w)}{\partial w} w = 0 \text{ for } l = 1, \ldots, L$$

in terms of elasticities

$$\sum_{k=1}^{L} \varepsilon_{lk}(p,w) + \varepsilon_{lw} = 0$$

If we increase all prices and wealth by a same percentage, demand does not change.

Two implications of Walras law

Walras law: $p.x(p,w) = w$

Cournot aggregation: The relation between own and cross-price effects

$$\sum_{l=1}^{L} p_l \frac{\partial x_l(p,w)}{\partial p_k} + x_k(p,w) = 0 \text{ for } k = 1, \ldots, L$$

Total expenditure cannot change in response to a change in prices.

Let $b_l = p_l x_l(p,w)/w$ be the budget share of the expenditure on good l

$$\sum_{l=1}^{L} b_l \varepsilon_{lk} + b_k = 0$$

Useful because we might have information about the elasticity of some goods, but we want to find out about the elasticity of others.

Two implications of Walras law

Walras law: $p.x(p,w) = w$

Engle aggregation: The relation between the income elasticity of several goods

$$\sum_{l=1}^{L} p_l \frac{\partial x_l(p,w)}{\partial w} = 1$$

Total expenditure must change by an amount equal to any wealth change.

In terms of elasticities

$$\sum_{l=1}^{L} b_l \varepsilon_{lw} = 1$$

Worked check: a two-good demand system

Let $0 < a < 1$ and

$$x_1 = \frac{aw}{p_1}, \qquad x_2 = \frac{(1-a)w}{p_2}.$$

- Budget shares are $(a, 1-a)$.
- Income elasticities are $(1, 1)$.
- Own-price elasticities are -1; cross-price elasticities are 0.

Verify homogeneity, Cournot aggregation, and Engel aggregation directly.

Numerical check: $a = 1/2$, $p = (2, 1)$, $w = 12$ gives $x = (3, 6)$.

WARP and the law of demand

Definition

Demand function $x(p, w)$ satisfies the WARP if

$$p.x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w), \text{ then } p'.x(p, w) > w'$$

Rephrasing of WARP:

If $p.x(p', w') \leq w \Rightarrow x(p, w)$ is revealed preferred to $x(p', w')$.

WARP says when both are affordable $x(p, w)$ must be preferred.

When $x(p', w')$ is chosen, $x(p, w)$ was not affordable.

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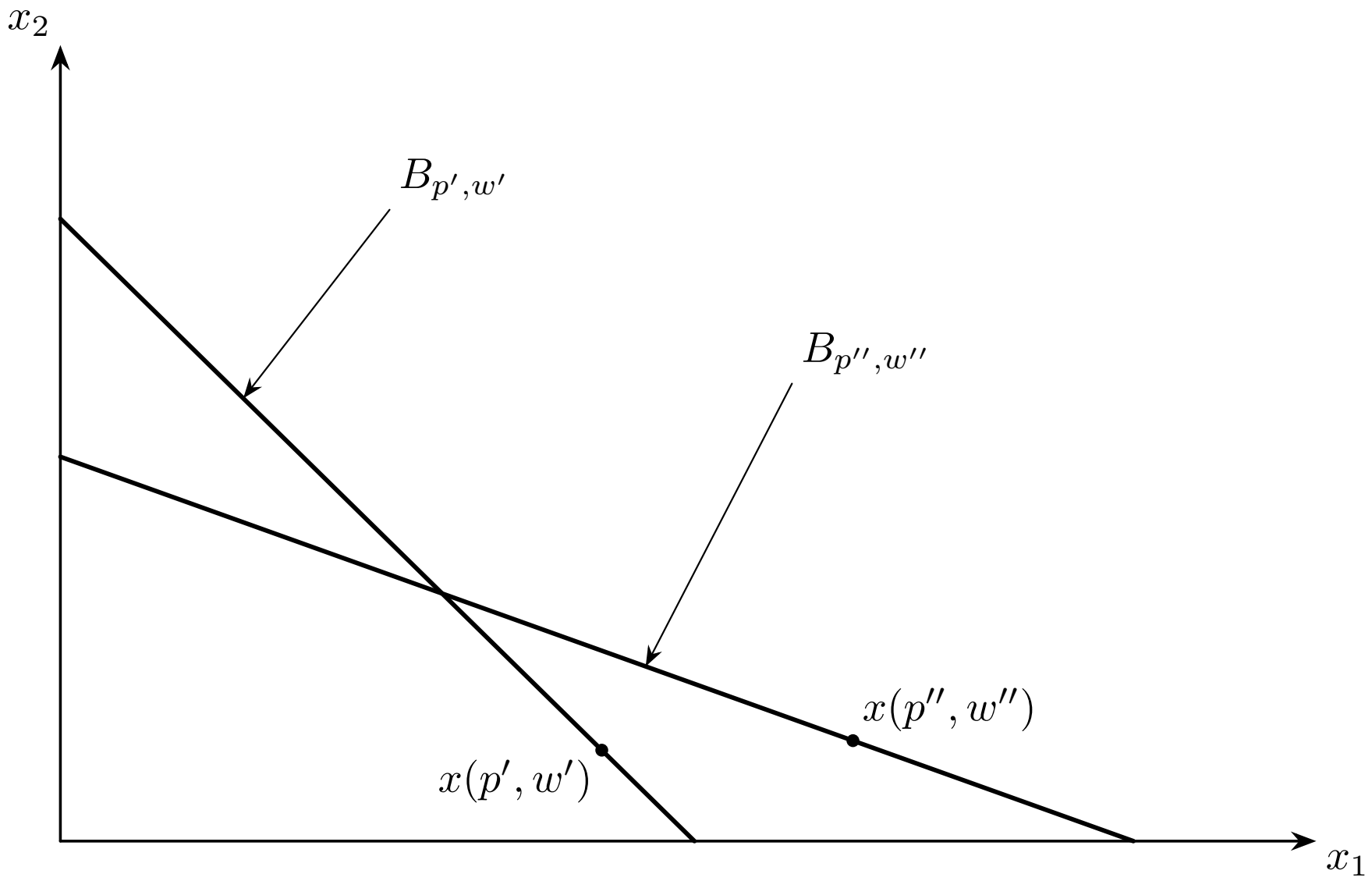
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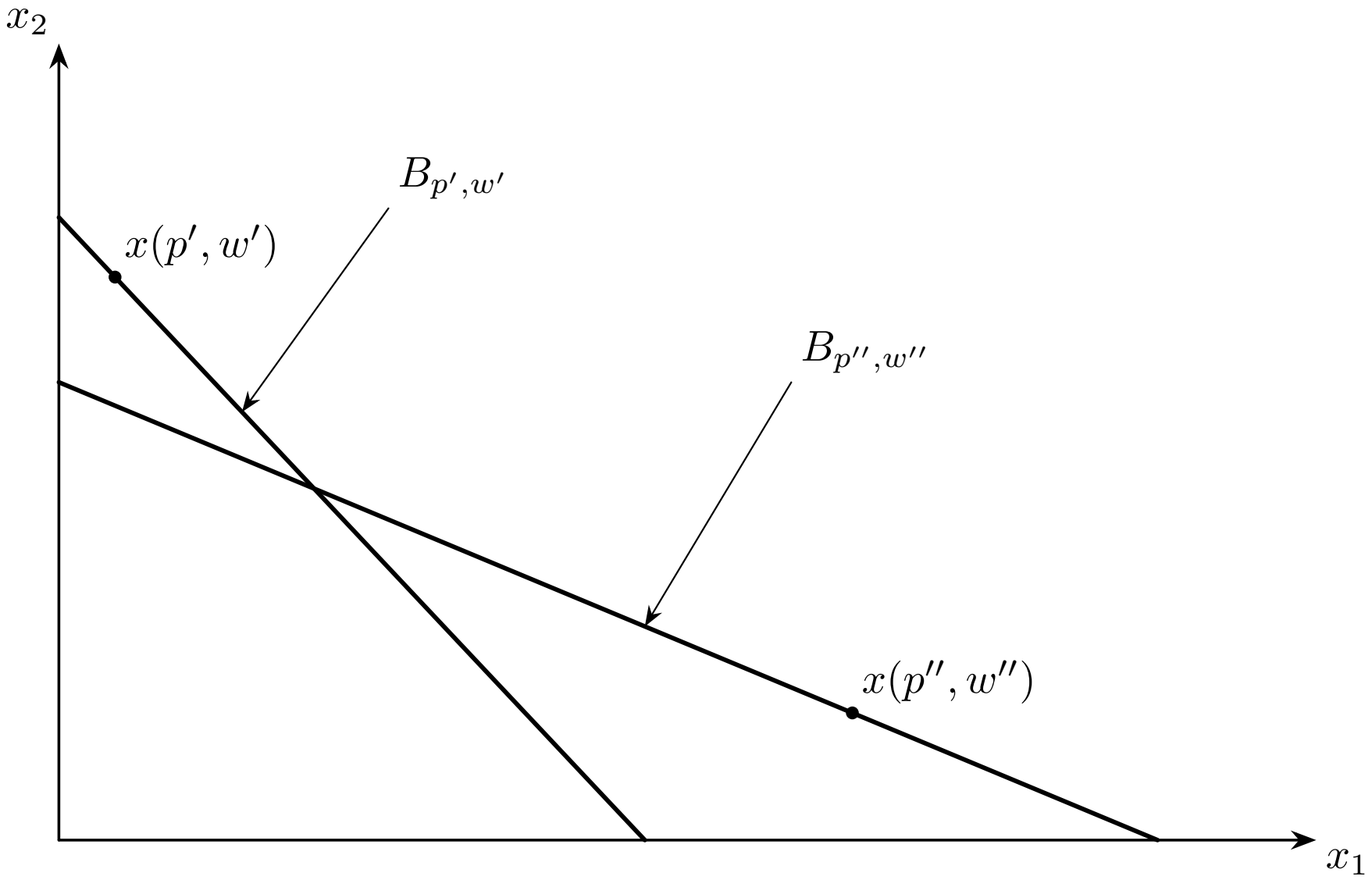
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What is the role of being a demand "function"?

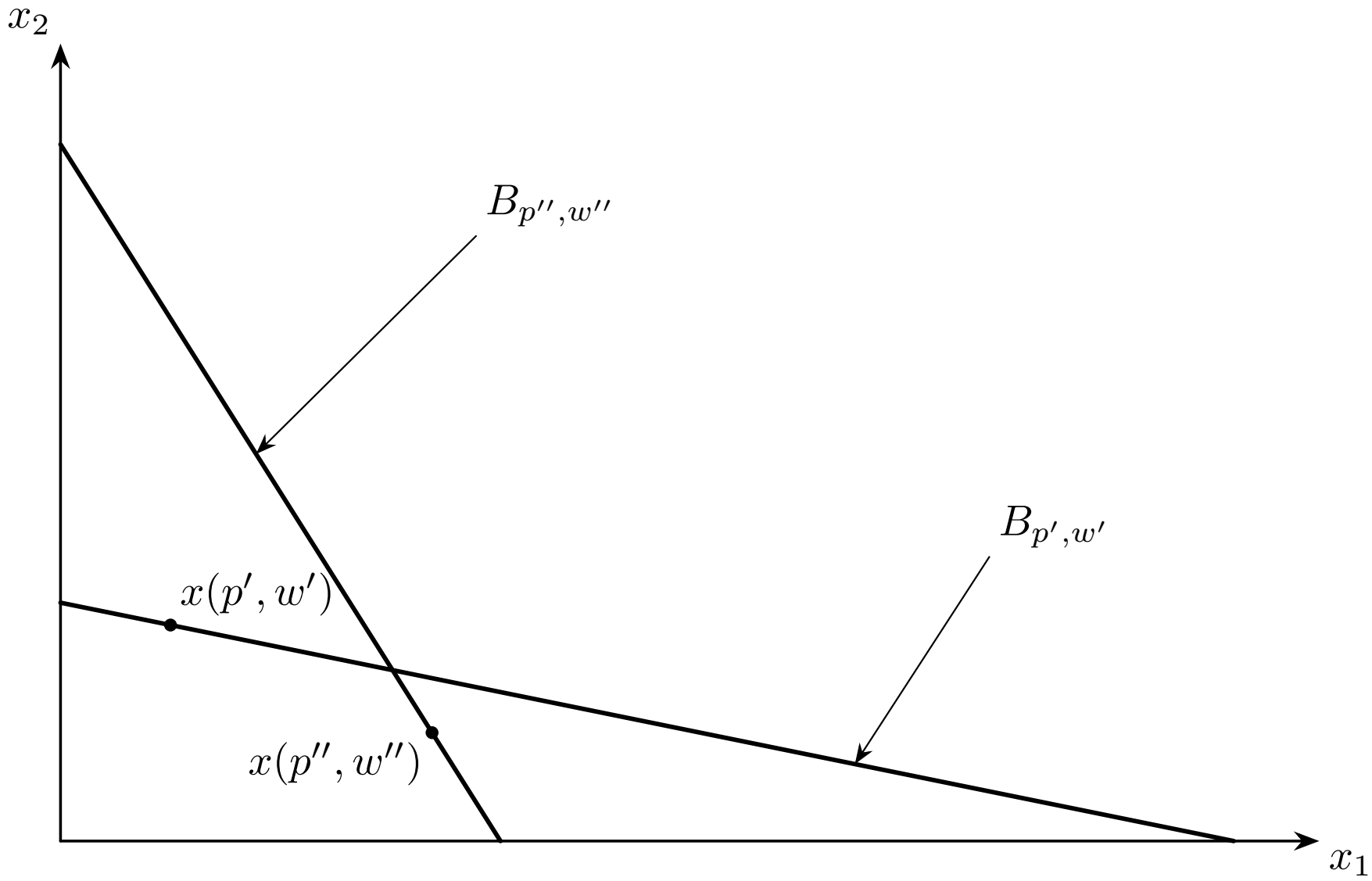
WARP examples



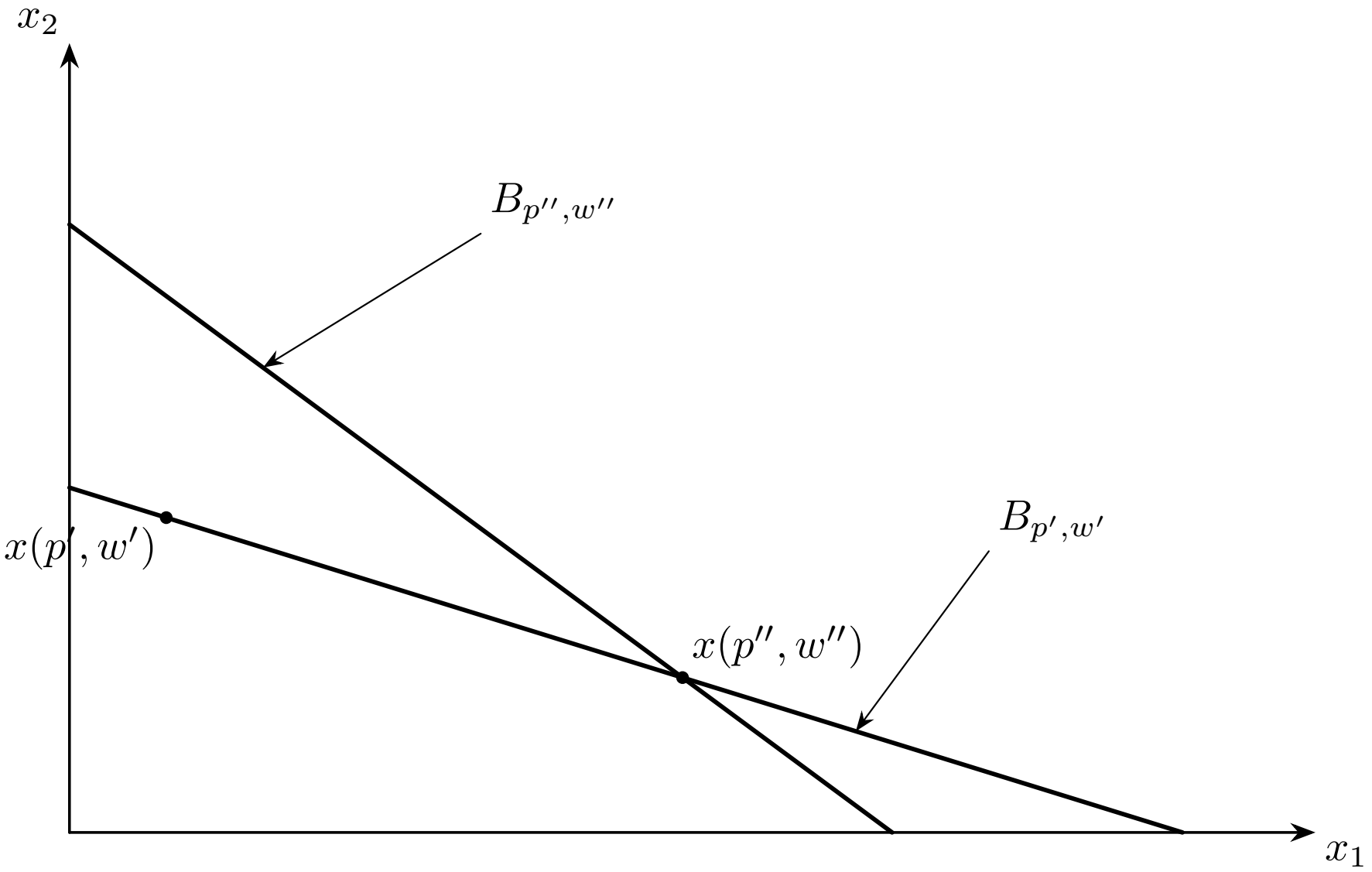
WARP examples



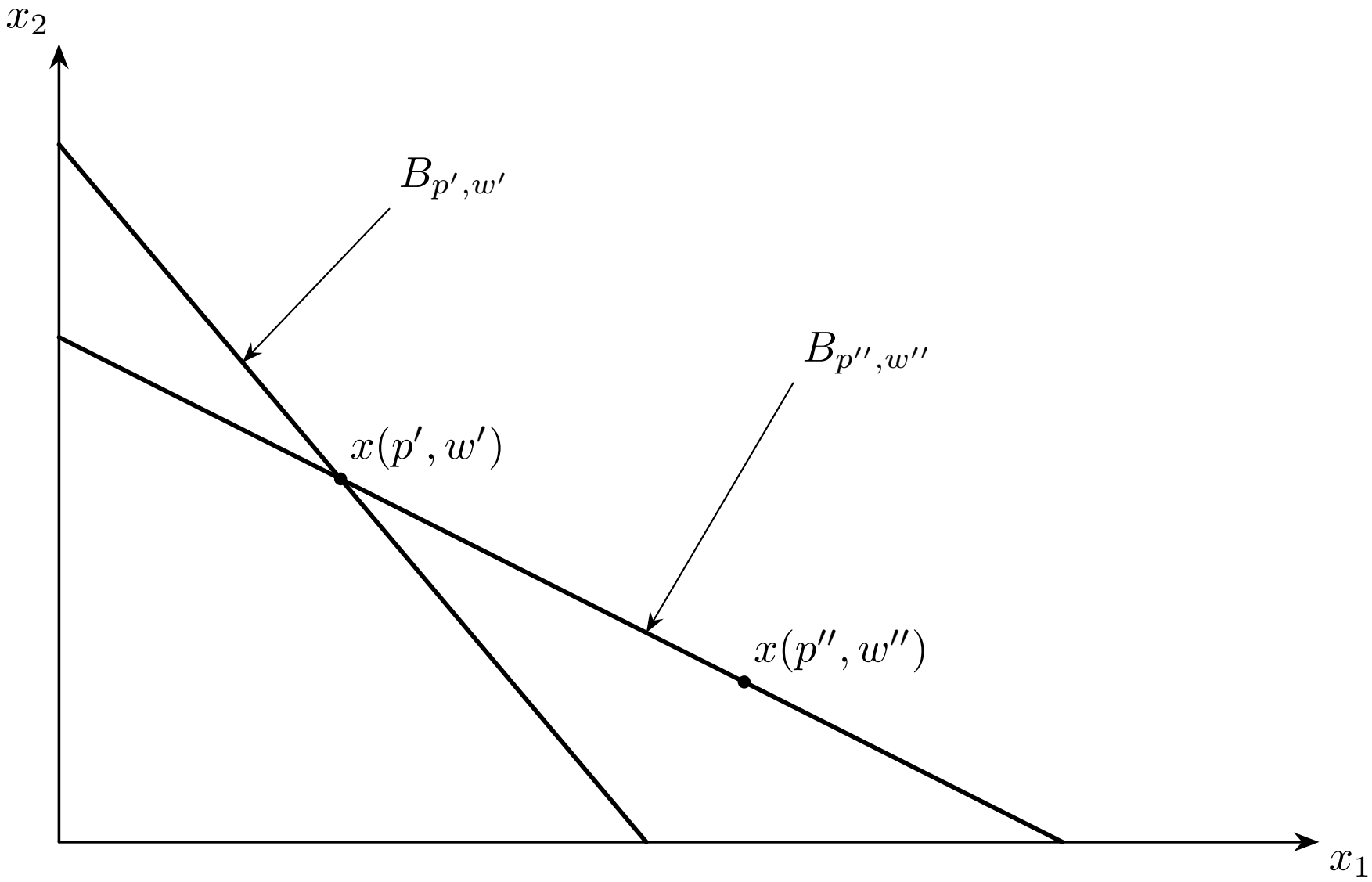
WARP examples



WARP examples



WARP examples



Compensated price changes

Slutsky wealth compensation: if p changes to p', we adjust wealth such that the previous bundle is just affordable: $w' = p'.x(p, w)$.

Proposition

Walrasian demand satisfies WARP if and only if:
For a compensated price change from (p, w) to $(p', w') = (p', p'.x(p, w))$, we have $(p' - p).[x(p', w') - x(p, w)] \leq 0$
with strict inequality whenever $x(p, w) \neq x(p', w')$.

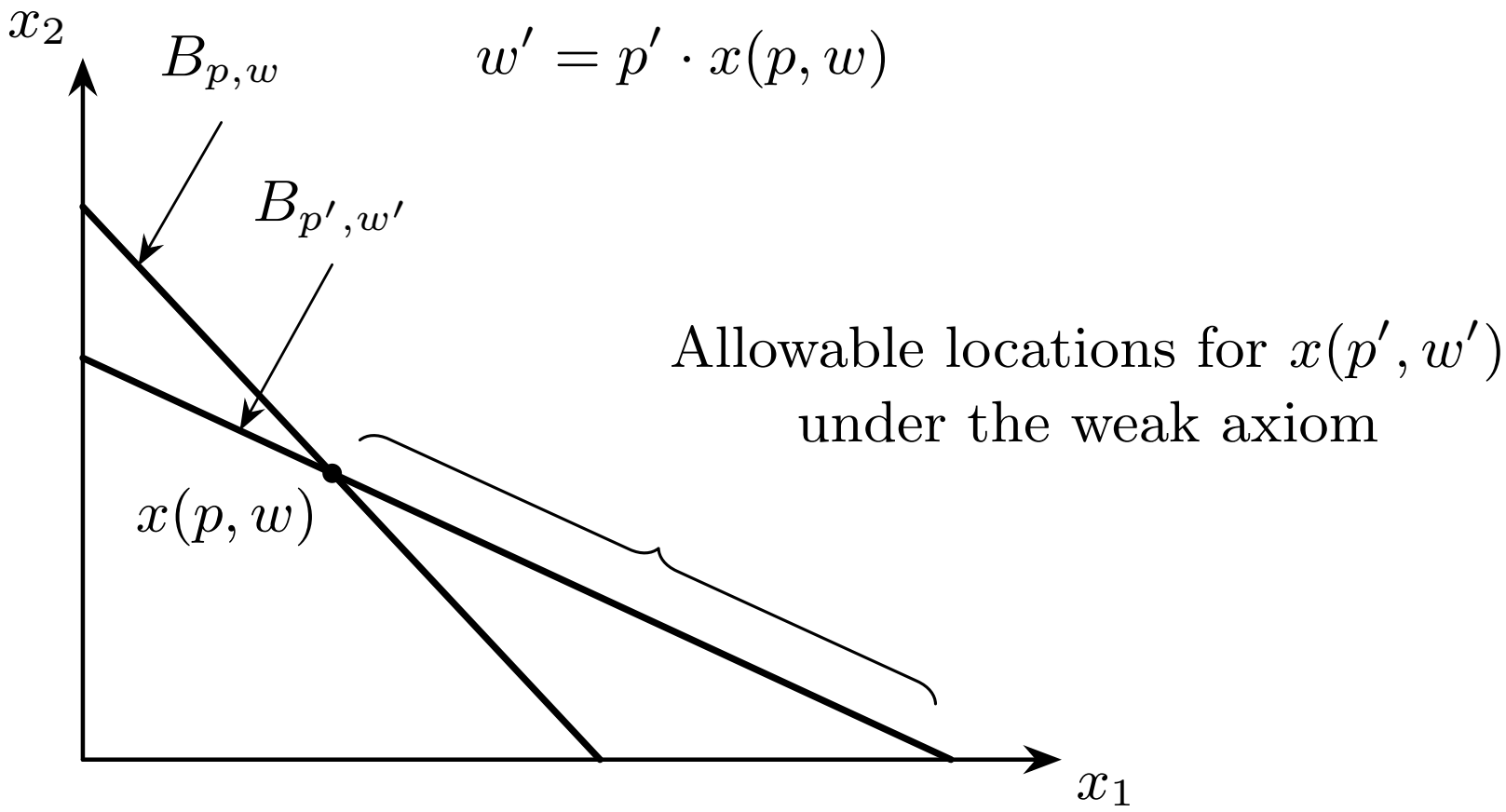
"Compensated law of demand": $\Delta p.\Delta x \leq 0$.

Proof.

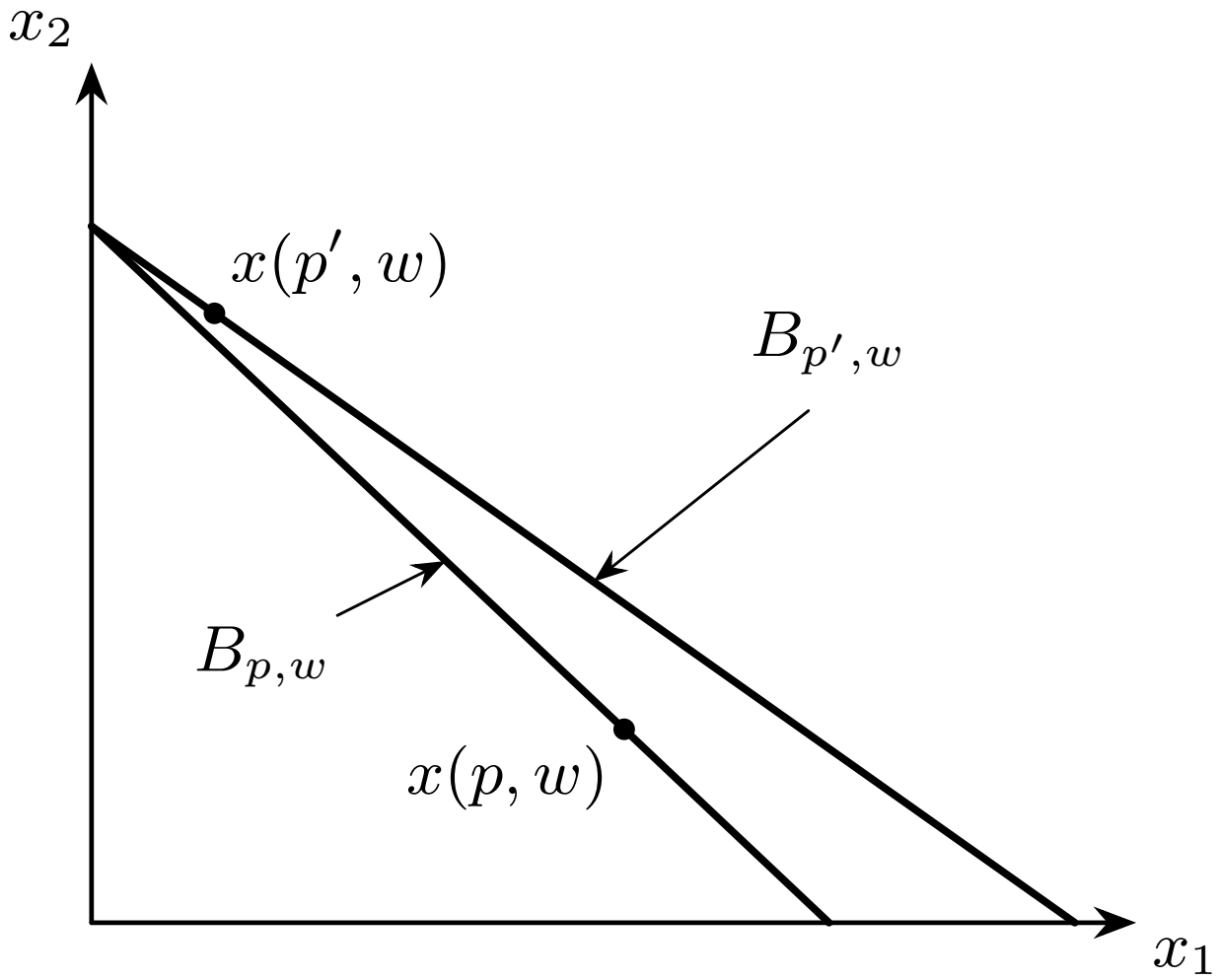
$$\Delta p.\Delta x = \underbrace{p'.[x(p', w') - x(p, w)]}_{\text{by Walras' law=0}} - \underbrace{p.[x(p', w') - x(p, w)]}_{by WARP>0}$$

□

Compensated price changes and WARP



Uncompensated price change



Slutsky matrix

using differentials for a compensated price change dp:

$$dx = D_p x(p,w)dp + D_w x(p,w)dw = D_p x(p,w)dp + D_w x(p,w)x(p,w)^T.dp$$

thus $dp.dx \leq 0$ is equivalent to

$$dp^T[D_p x(p,w) + D_w x(p,w)x(p,w)^T]dp \leq 0$$

Proposition

If a differentiable Walrasian function satisfies WARP, then at any (p,w) the Slutsky matrix $S(p,w)$ is negative semidefinite
($\forall v \in \mathbb{R}^L : \; v^T S(p,w) v \leq 0$)

With the results so far Slutsky matrix is not necessarily symmetric.

Slutsky matrix

Slutsky matrix elements:

$$s_{lk}(p,w) = \frac{\partial x_l(p,w)}{\partial p_k} + \frac{\partial x_l(p,w)}{\partial w} x_k(p,w)$$

Proposition

For differentiable Walrasian demand functions
$p^T S(p,w) = S(p,w)p = 0$

In sum:

- WARP is equivalent to compensated law of demand
- The compensated law of demand implies negative semidefiniteness of Slutsky matrix